

## **LAB 3 – ARCHITECTURAL JUSTIFICATION**

### **Self-Service Coffee Kiosk System**

#### **Architecture Selection**

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

#### **Reason 1 – Separation of Concerns**

Layered Architecture separates the touchscreen user interface, order management, payment processing, receipt printing, and data storage into distinct responsibilities. This makes the system easier to understand and maintain.

#### **Reason 2 – Simplicity**

The coffee kiosk has a relatively straightforward ordering workflow, so a layered structure provides a clear organization without introducing the operational complexity associated with microservices.

#### **Security Advantage**

The layered structure separates the user interface from payment and data-handling components, helping restrict direct access to sensitive payment and database functionality.

#### **Performance Benefit**

The architecture keeps the kiosk's ordering operations organized into focused components, allowing requests such as menu retrieval, payment processing, and receipt generation to follow clear component interactions without the additional network overhead associated with a distributed microservices architecture.